

Where You Refuel — Technical Report

Dataset description, exploratory analysis and data quality assessment

Data Visualization · Deliverable 2

MIMIT Osservaprezzi Carburanti · Nord-Est Italy · 2024-2025

1. Scope and purpose

This report documents the data behind a single question — *how much does where you refuel change what you pay?* — restricted to the 22 provinces of ISTAT NUTS-1 **Nord-Est** (Trentino-Alto Adige, Veneto, Friuli-Venezia Giulia, Emilia-Romagna) over **2024-2025**, sampled **one day per week (Monday)**.

The restriction is deliberate. The question is geographic, not dynamic, so daily resolution buys nothing; and, as Section 5.3 shows, the 2024-2025 window is internally consistent in a way the full 2015-2026 archive is not.

Headline: 105 weekly extracts yield **2,000,810 usable price observations** (97.14% of the Nord-Est rows) across **5,329 registered stations**, with only 125 rows discarded for quality reasons. The dataset is in good condition; its real difficulties are semantic (free-text fuel names, mixed units) rather than structural.

2. Sources and provenance

Source	Content	Licence
MIMIT <i>Carburanti</i> — <i>archivio storico prezzi</i>	daily station prices, quarterly .tar.gz, 2015 Q1-	IODL 2.0
MIMIT <i>Carburanti</i> — <i>anagrafica impianti attivi</i>	station registry with coordinates	IODL 2.0
openpolis/geojson-italy	province boundaries (ISTAT-derived)	CC-BY

Prices are reported to the Ministry by operators under art. 51 of Law 99/2009. Each daily file is an extract of the prices valid at 08:00. Archive URLs follow `opendatacarburanti.mise.gov.it/categorized/{dataset}/{year}/{year}_{q}_tr.tar.gz`.

Verification. Five of the 105 weekly price files were re-extracted from freshly downloaded quarterly archives and compared by MD5 against the working copies. All five matched byte-for-byte.

3. Structure of the raw data

Prices (`prezzo_alle_8-YYYYMMDD.csv`) — line 1 is a free-text extraction stamp, line 2 the header, delimiter `;`:

Field	Type	Notes
idImpianto	int	joins to the registry
descCarburante	text	free text , see §5.4
prezzo	float	EUR/litre, or EUR/kg for methane and LNG
isSelf	0/1	1 = self-service, 0 = attended
dtComu	datetime	when the operator filed this price

Registry (`anagrafica_impianti_attivi-YYYYMMDD.csv`) — same two-line preamble, delimiter `;`, ten fields: `idImpianto`, `Gestore`, `Bandiera`, `Tipo Impianto`, `Nome Impianto`, `Indirizzo`, `Comune`, `Provincia`, `Latitudine`, `Longitudine`. `Tipo Impianto` is a clean binary in this period: `Stradale` or `Autostradale`.

A national daily file holds ~93,000 price rows across ~24,000 stations. Eight registry snapshots were taken, one per quarter, growing from 22,821 to 24,462 national rows over the two years.

4. Construction of the analysis panel

The pipeline (`scripts/`, ~600 lines of Python) runs in five stages:

1. **geo.load_registry** — parse all eight snapshots, restrict to Nord-Est, derive region, motorway flag, harmonised brand, and a geocode validity flag; take the union across snapshots keeping the most recent record per station.
2. **build_panel** — read the 105 weekly price files, join the registry, harmonise fuel labels, apply the documented filters, and write the panel plus a machine-readable QC record (`data/processed/qc.json`).
3. **analyse** — compute the four geographic views.
4. **figures / wireframe** — render.

Two methodological rules are applied throughout, because both guard against composition effects that would otherwise be mistaken for geography:

- Statistics are computed **within a week** and then averaged across weeks. Averaging raw prices over two years would let a province that happens to report more often during expensive weeks look expensive.
- Paired comparisons (the attended premium) are made **within a station**, so the gap cannot be an artefact of *which* stations offer attended service.

5. Data quality assessment

5.1 Coverage

All 105 expected Mondays are present; none is missing. Between 4,301 and 4,439 Nord-Est stations post at least one price in any given week, out of 5,329 registered. Over the whole period **5,018 stations report at least once**, so **311 registered stations (5.8%) are never observed quoting a price** — they are listed as active but silent. This is a real limitation: the “provincial mean” is a mean over stations that report, not over stations that exist.

5.2 Price plausibility

Only **124 rows (0.006%)** fall outside a plausibility band of 0.5–4.0 EUR/litre and are removed; one further row has an unparseable station id. Nothing is missing: `prezzo` parses for every remaining row. The retained distribution is well-behaved, with pronounced spikes at round numbers (1.700, 1.800, 1.900) that are a genuine pricing behaviour rather than a defect.

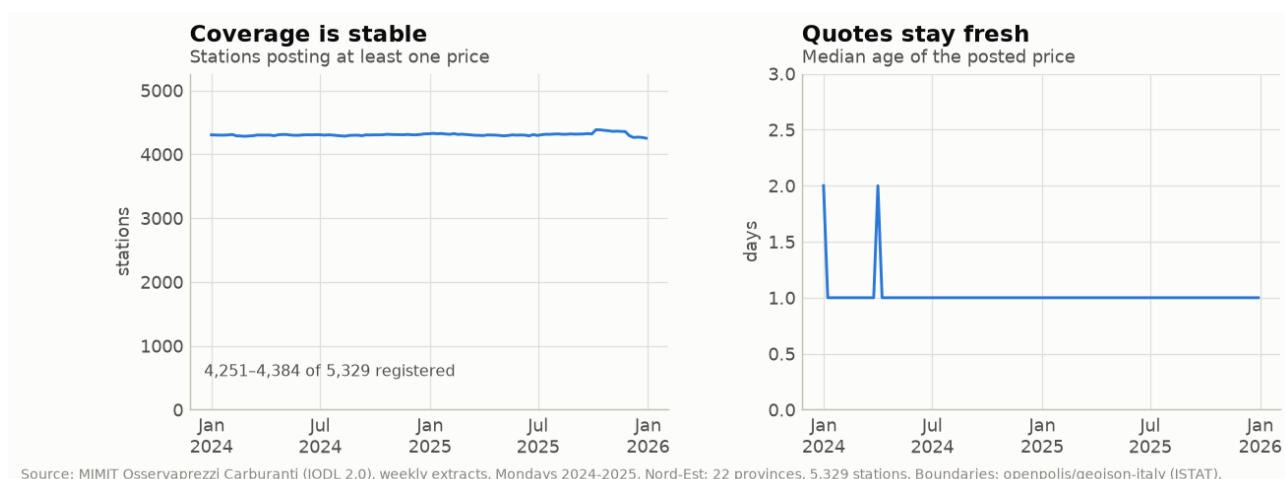


Figure 1: Weekly coverage and the freshness of quotes.

For context, the *live* feed downloaded on 2026-08-04 carries values of 0.100 and 8.888 — placeholders rather than prices. The band is set to catch exactly these without touching any real fuel price.

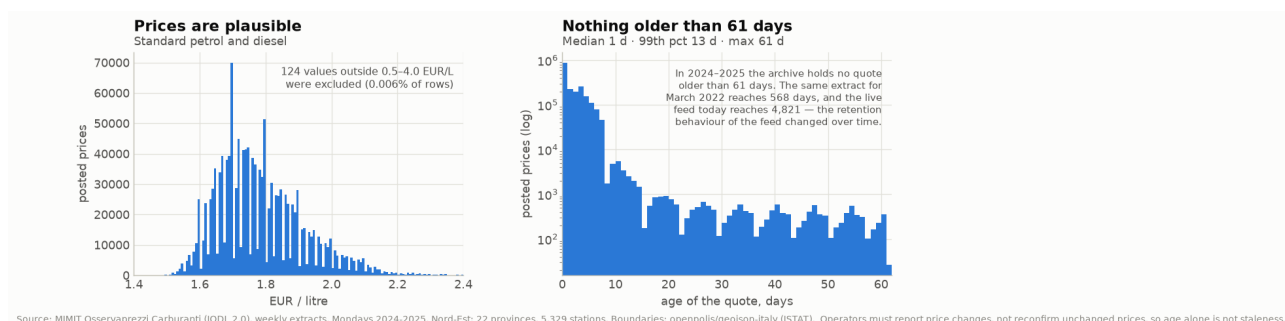


Figure 2: Price plausibility and the age of quotes.

5.3 Quote age — and a regime change in the archive

Operators are required to report price *changes*, not to reconfirm unchanged prices, so an old dtComu is **not by itself evidence of a stale record**. We therefore flag at 60 days and drop only beyond 365.

In this window the filter never fires. Median quote age is **1 day**, the 95th percentile 6 days, and the maximum **61 days**. No row is older than that.

That cleanliness is a property of *this window*, not of the dataset:

Extract	max age	99th pct	rows > 60 d
Archive, 2022-03-17	568 d	453 d	6,201
Archive, 2024-2025 (this panel)	61 d	13 d	0
Live feed, 2026-08-04	4,821 d	19 d	235

MIMIT's retention behaviour evidently changed between 2022 and 2024, and the live feed behaves differently again. **Any project spanning the full archive would have to model this**

discontinuity; scoping to 2024-2025 avoids it, and that is part of why the window was chosen.

One caveat: the age comparison must be made date-to-date, not timestamp-to-timestamp. The 08:00 extract carries quotes filed up to 08:15 the same morning, so a naive comparison against midnight scores 43% of rows as “filed in the future”. This was an early bug in our pipeline, caught because the implied exclusion rate was implausible.

5.4 Fuel labels: the main semantic problem

descCarburante is free text. Across the study period it takes **59 distinct values** (56 after case-folding) for what are really **nine products**. Diesel alone appears under 20 commercial names — Blue Diesel, Hi-Q Diesel, Supreme Diesel, Excellium Diesel, Gasolio Oro Diesel, and so on.

Harmonisation is an explicit lookup table (fuels.py), not a fuzzy rule, so an unrecognised label surfaces as an error rather than being silently absorbed. After harmonisation **zero rows are unmapped**.

Two decisions matter for the analysis:

- **Premium grades are kept separate from standard ones.** Pooling them would make a station that happens to sell Blue Diesel look expensive when it is simply selling a different product.
- **Winter-grade diesel** (Gasolio artico, Gasolio Alpino, ...) is separated too: it is sold mainly in Alpine provinces and priced above standard diesel, so pooling it would bias precisely the provinces being compared.

Mixed units. Methane, LNG and L-GNC are priced **per kilogram**, not per litre. These **58,784 rows (2.854%)** are excluded from every EUR/litre view. This is the single largest exclusion in the pipeline, and it is a unit incompatibility rather than a quality failure.

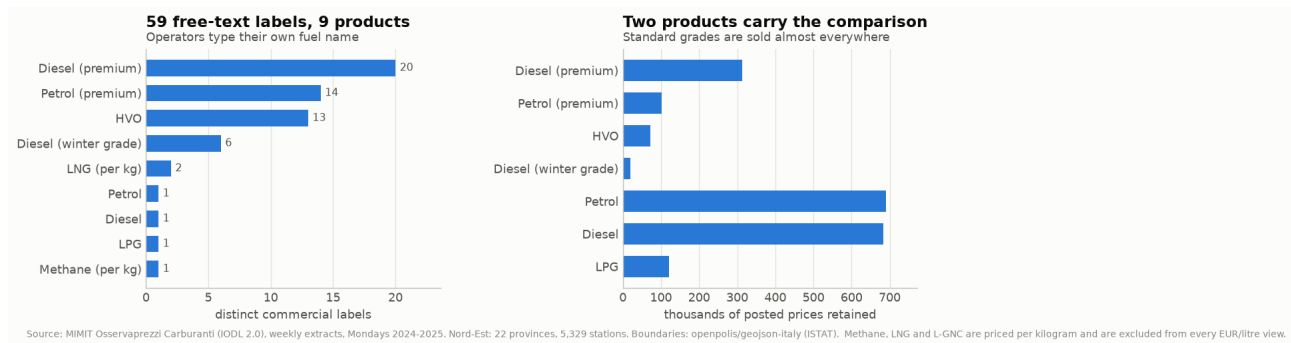


Figure 3: Label harmonisation and the products that carry the comparison.

5.5 Registry quality

- **Embedded separators.** 34 rows across the eight snapshots carry a stray ; inside the station name or address. Because idImpianto is always first and the coordinates always last, these are *recoverable* by anchoring from both ends; we repair them rather than drop them. **Zero rows are unrecoverable.**
- **Geocoding.** Only **10 stations (0.19%)** have coordinates that fall outside a Nord-Est bounding box or sit at (0,0). They are excluded from spatial views.
- **Churn.** 4,267 of 5,329 stations appear in all eight snapshots; 225 appear in only one. Because a single snapshot cannot resolve every id, the registry is built as a union across snapshots — using only the latest snapshot would have silently dropped historical stations.

- **Rebranding.** 58 stations change Bandiera during the period. Brand is therefore a property of the *latest* record, and brand-level comparisons should be read with that in mind.

5.6 Exclusion ledger

Filter	Rows	% of Nord-Est rows
Unparseable station id	1	0.000
Price missing	0	0.000
Price outside 0.5–4.0 EUR/L	124	0.006
Fuel label unmapped	0	0.000
Priced per kilogram (methane, LNG)	58,784	2.854
Service flag unknown	0	0.000
Quote older than 365 days	0	0.000
Retained	2,000,810	97.14

6. Exploratory analysis

The panel covers 5,008 geolocated stations that reported at least once — 101 of them on the motorway. Composition: Veneto 2,241 stations, Emilia-Romagna 2,065, Friuli-Venezia Giulia 587, Trentino-Alto Adige 436; by brand, Eni 1,153, IP 751, unbranded *pompe bianche* 622, Q8 562, Esso 496, Tamoil 324, Shell 26, and 1,395 others.

All figures below use **standard self-service grades**, the products nearly every station sells and therefore the only ones comparable across 5,000 pumps.

6.1 Between provinces

Mean self-service petrol ranges from **€1.745 in Rovigo** to **€1.833 in Bolzano** — a spread of **8.8 cents**. Diesel behaves similarly (€1.660 to €1.758, 9.8 cents). The pattern is coherent rather than noisy: Alpine and border provinces (BZ, TN, TS, GO) are dear, the Po plain (RO, TV, PD, FE) is cheap.

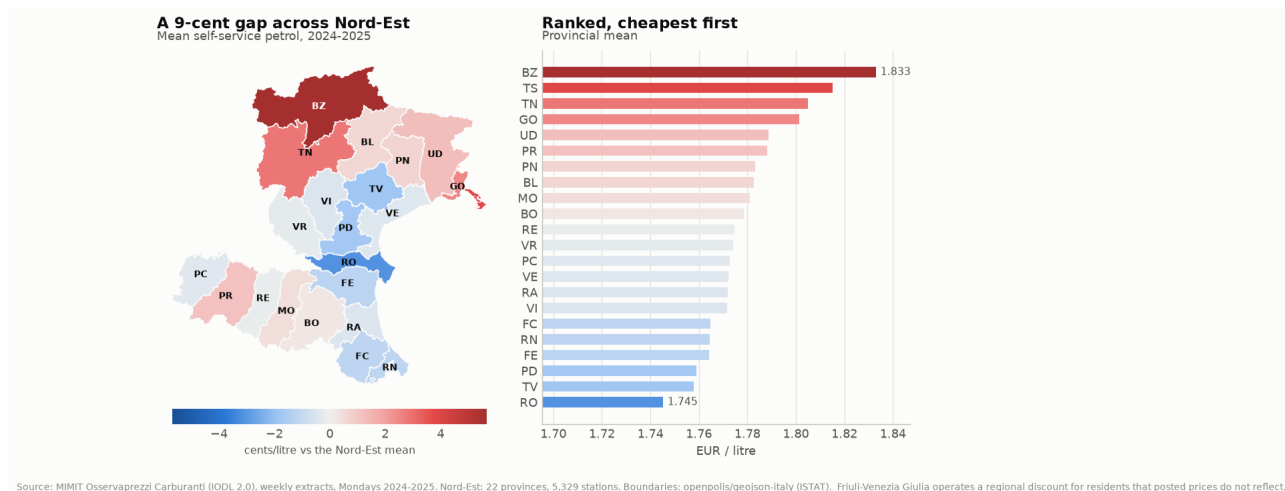


Figure 4: Provincial deviation from the Nord-Est mean.

6.2 Motorway

The motorway premium averages **10.4 cents** for petrol and **11.6** for diesel, and never closes: the weekly minimum across two years is 7.4 cents. The two populations are almost disjoint in the distribution plot. Note the small sample — 101 reporting motorway stations against ~4,900 ordinary — which is why medians are used throughout.

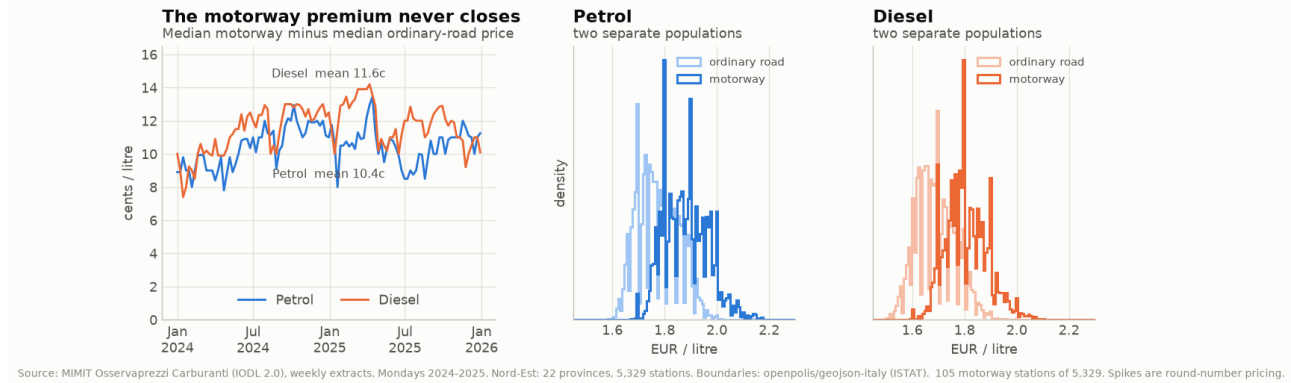


Figure 5: The motorway premium, and the two price populations.

6.3 Within provinces — the result that reframes the question

Within a single province and a single week, the 10th-90th percentile of station prices spans **9.1 cents on average** — *as wide as the entire 8.8-cent range between the cheapest and dearest province*. Bologna is the most dispersed (11.8c), Bolzano the least (6.0c): Bolzano is uniformly expensive, whereas Bologna contains both cheap and dear pumps.

The consequence for the visualization is direct. A choropleth alone would leave the reader with “my province is expensive”; the data supports “my province contains both the cheapest and the dearest pump I could reach”, which is both more accurate and more actionable.

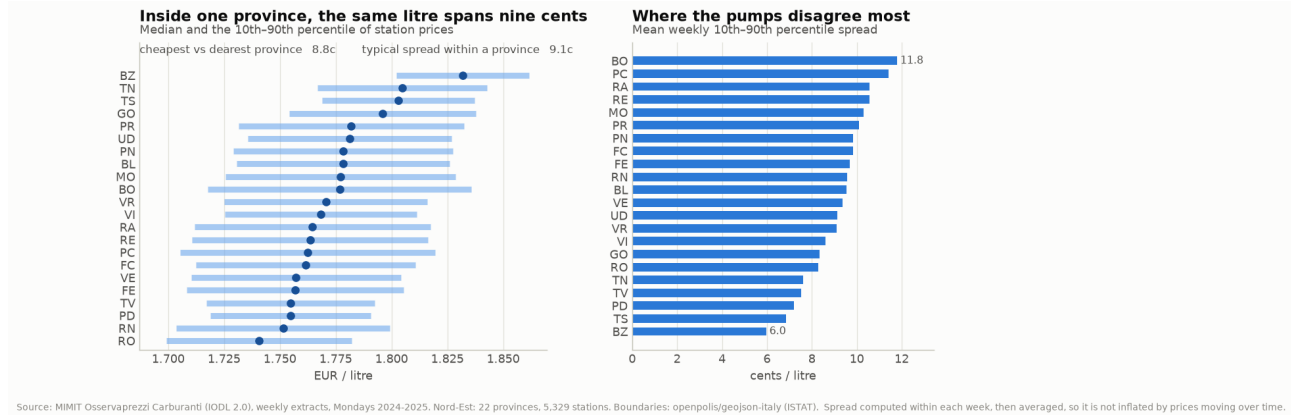


Figure 6: Within-province dispersion.

6.4 Attended versus self-service

Measured within station and week, across **240,347 paired station-weeks at 2,785 stations**, attended service costs **13.0 cents** more on average — larger than any other gap in the study.

That mean, however, describes almost no real station. The distribution is strongly **bimodal**: **32% of paired station-weeks show no differential at all**, while among those that do the gap averages **19.0 cents**. Roughly 30% of stations never differentiate. Reporting only the mean would misrepresent both groups, so the figure shows the shape.

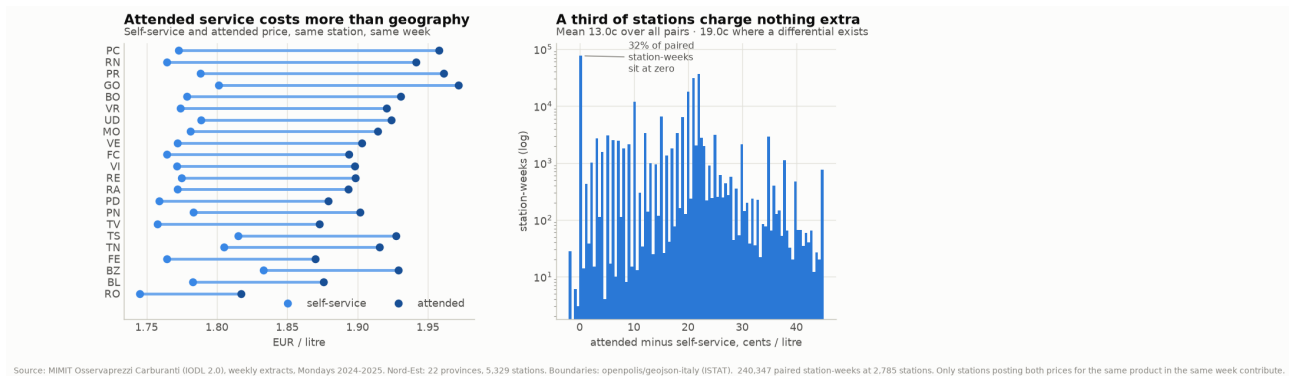


Figure 7: The attended-service premium.

6.5 Synthesis

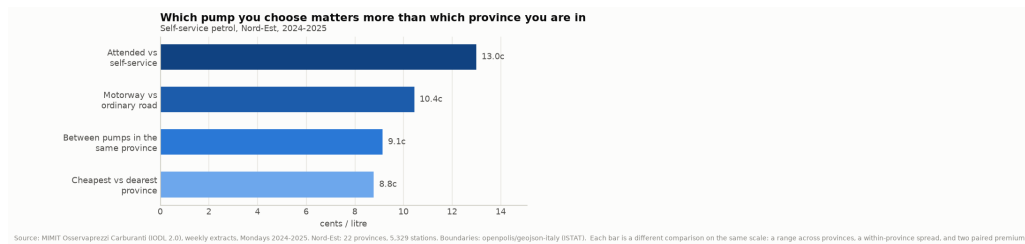


Figure 8: The four gaps on one scale.

Ordered by size, the four gaps a driver faces are: attended service 13.0c, motorway 10.4c, within-province spread 9.1c, and between-province range 8.8c. **Geography matters least.**

7. Limitations

1. **Posted, not transacted, and unweighted.** Prices are what operators advertise, not what drivers pay, and every station counts equally regardless of throughput. Provincial means are means over pumps, not over litres.
2. **Silent stations.** 311 registered stations never report; if silence correlates with price, provincial means are biased.
3. **Friuli-Venezia Giulia** operates a regional discount for residents that posted prices do not reflect, so UD, GO, TS and PN overstate what a resident actually pays.
4. **Weekly Monday sampling** is appropriate for a geographic question but would be inadequate for a dynamic one.
5. **Brand is time-collapsed** to the latest snapshot, affecting the 58 rebranded stations.
6. **The clean age profile does not generalise** beyond 2024-2025 (\$5.3).

8. Reproducibility

scripts/config.py
scripts/fuels.py

scope, paths, thresholds
59 labels -> 9 products; per-kg products flagged

<code>scripts/geo.py</code>	registry parsing, repair, geocode QC, boundaries
<code>scripts/build_panel.py</code>	105 weekly files -> panel + qc.json (~95 s)
<code>scripts/analyse.py</code>	the four geographic views
<code>scripts/figures.py</code>	figures 01-08
<code>scripts/wireframe.py</code>	figure 09

Running `build_panel.py`, `analyse.py`, `figures.py` and `wireframe.py` in that order regenerates every number and figure in this report from the raw CSVs. Every threshold is a named constant in `config.py`; every exclusion is counted into `data/processed/qc.json` rather than applied silently. Environment: Python 3.12, pandas 3.0.5, matplotlib 3.11.1, numpy 2.5.1, pyarrow 25.0.0.

The chart palette is the validated reference palette from the course's visualization guidance; the two-series categorical pair was checked programmatically for colour-blind separation on the light surface (worst adjacent CVD ΔE 24.7 protan, normal-vision ΔE 33.6) rather than by eye.